

Watcher v2 — Build Roadmap

From a message filer to a receptionist. Every item scored for urgency and ease. • **v2.13** • 23 August 2026 • supersedes the v2.12 PDF

What this revision does. v2.11 delivered 2.4; v2.12 reconciled B5's durable queue onto main and re-opened B4's webhook claim after a direct check found zero traffic. **This revision closes what those left open, verified against the live systems.** First, **B5 is now fully done:** the missing piece — the arq worker service — exists as **watcher-worker**, a paid Render Background Worker, deployed and live on current main, its own logs confirming it started for `consume_message` and connected to Redis; `watcher-redis` backs it. Second, **B4 is done bar one item:** the Meta webhook subscription and the operator number are completed operator-side, closing the two checks session 11 re-opened, and the custom domain and both Meta credentials remain confirmed — the single sub-item still open is moving `watcher-api` off the Render free plan, so a cold start no longer reads to Meta as a timeout. Third, migration **005 (the facts table) was found still unapplied to production and applied** this session — the live schema is now `005_facts`, so `answer_from_knowledge` no longer risks a missing-table error on the first real question. Fourth, **Track 2 is complete:** 2.8 (many properties per client) — a properties table, per-property fact scoping, message → property resolution, 521 → 535 tests — and 2.7, which re-recorded the eval live under prompt v3 and moved the baseline from v2's 0.88 to a measured **0.98**. **Remaining: ~28.0 engineering days.**

Where we stand today

Built and tested	Missing — nothing ships without these
535 passing tests, no DB or network needed 91 source files across 20 modules (133 with tests) 21 DB tables + 6 migrations — through 005 applied to the live database; 006 (properties) applies on deploy Every external system behind a swappable seam Eval runner + CI accuracy gate, 56 golden cases, v3 baseline 0.98 Receptionist vocabulary: 19 intents, data not code Prompt v3 — catalogue rendered from the vocabulary Channel-neutral envelope + chat/voice adapters Task state machine and the autonomy gate Typed config over every .env variable (A1) Anthropic + OpenAI on the LLM seam, caching on (A3) Engine + session scope, pooler-safe by default (A2) Composition root: the process starts and files a message end to end (A4) Conversation continuity: one task carried across turns, classifications recorded (A5) Outbound sender — the reply reaches the guest (A6) Supabase project, migrated, with the channel_configs row (B1) RLS forced on every table, per-transaction tenant GUC, cross-tenant test (B2) Container image; the project installs from pyproject (B3) Deployed and serving on Render, Frankfurt (B3) Emergency detection — the declared triggers matched before the classifier, in two scripts and Franco-Arabic (G3) The alert path — an operator seam, an implementation, and an honest report of which channel was used (G3) Knowledge base — a facts table behind RLS, fuzzy-matched against a guest's own words, a sensitivity flag a fact can carry, and a real "I don't know" (2.4) Durable queue — a fourth ClassificationQueue transport on arq/Redis, an arq worker as its own composition root, and one shared graph-builder so the worker and the API can't drift (B5) NEW • Deployed and consuming — watcher-worker (a paid Render Background Worker) and watcher-redis both live; the worker's logs confirm it started for consume_message and connected to Redis (B5) NEW • facts table live in production — migration 005 applied to the Supabase DB this session, RLS forced, schema now at 005_facts (2.4)	No DB connection — done (A2) No endpoint — done (A4) No memory across turns — done (A5) No outbound — done (A6) No emergency detection — done (G3) No slot extraction — the model emits no slots, so a task fills only by handing off (2.x) No database — done (B1) No RLS — done (B2) No Dockerfile — done (B3) No running service — done (B3) Knowledge: zero tables, zero rows — table live, migration 005 applied (2.4) No outbound credentials — set on Render, confirmed by a clean redeploy, not by a test send (B4) No worker service — watcher-worker live, a paid Background Worker consuming (B5) No operator number — CONTROL_CHAT_PHONE_E164 completed operator-side (B4) No webhook subscription — completed operator-side; first inbound message is the proof to watch for (B4) Still on the free Render plan — the one B4 item left: watcher-api sleeps after ~15 minutes and a 30–60s cold start reads to Meta as a timeout, so the warm-instance half of B4 needs the paid upgrade Narrow trigger phrases — the detector matches what an operator wrote down and does not paraphrase; "I smell gas" matches neither declared gas phrase. One line of YAML, and it is the operator's line The database path is unproven — SQLAlchemy connects lazily, so the pooler host is tested by the first message, not by startup No control page — one CSS file, no package.json No control-page API — ~25 endpoints, none written Live availability — no read path to any PMS A measured prompt — the gate replays v2 fixtures, so 88% is not v3's number

What changed in this revision (v2.12 → v2.13)

Delivered	Consequences
B5 fully done (session 12) — the worker service, the one piece open at the end of session 11, now exists. watcher-worker is a paid Render Background Worker running arq apps.api.worker.WorkerSettings, deployed and live on current main; its logs confirm it started for consume_message and connected to Redis. Verified via the Render MCP B4 done bar the plan upgrade — the Meta webhook subscription and the operator number (CONTROL_CHAT_PHONE_E164) are completed operator-side, closing the two checks session 11 re-opened. The custom domain and both Meta credentials stay confirmed Migration 005 applied to production — found still at 004 this session and brought to 005_facts: the facts table now exists behind forced RLS, with the tenant-isolation policy and the watcher_app grant, so answer_from_knowledge can no longer hit a missing table on the first real property_question 2.8 delivered (session 12) — a properties table (migration 006), per-property fact scoping, and message → property resolution 2.7 delivered — the eval was re-recorded live under prompt v3 (first-pass claude-haiku-4-5), moving the baseline from v2's 0.88 to a measured 0.98 (55/56); six Franco-Arabic golden cases added; scripts/measure_prompt.py and scripts/record_fixtures.py are the committed tools. Track 2 is complete Track B: 0.5 → 0.25 days — B5 done, B4 down to the paid-plan upgrade. With 2.7 and 2.8, total 29.75 → 28.0 Tests 521 → 535 (2.8's properties, resolver and scoped search); the eval gate now enforces the v3 0.98 baseline	M1 — answers a real message, safely — is within one operator action: the paid Render upgrade. Everything else on B4 is reported done; the first real inbound message producing a messages row (or a POST /webhook 200 in the logs) is the positive proof still worth watching for, since at check time both were still empty The queue is real end to end now, not just in tests. A message the API enqueues to Redis will survive a deploy and be picked up by watcher-worker — provided REDIS_URL is set on watcher-api to switch it to the producer branch, which stays a deliberate confirm-the-worker-consumes-first step A sensitive fact is narrowly gated, not fully gated. An unverified match to a sensitive fact is refused — the same “I don't know” as a genuine miss — but that is one tool refusing to guess, not G1's reply-path-wide disclosure gate, still open A client can be an agency now, not one flat. A fact scopes to a unit or stays tenant-wide (2.8); a guest on a shared number gets the tenant-wide facts, never another unit's, until a per-message property signal exists — the endpoint hint is wired and tested, the booking-based one waits on 3.1. Migration 006 applies on deploy, like 005 Unchanged: 3.1 still blocked on P1; slot extraction (item 2.x) is still unwritten; the eval still replays v2 fixtures, which 2.7's re-record is what changes — and it needs a key

Totals: ~28.0 engineering days remaining. At the observed 2–3 engineering days per working session that is ~10–14 sessions; at the six-day week these estimates assume, ~4.7 weeks of pure build — so plan ~7 weeks to sellable. The critical path is **B4's last item alone** — the paid Render upgrade — which is operational, not engineering; with the subscription, the operator number and the worker all reported or verified done, M1 is one billing action away.

What is left in front of a pilot. 2.4 closed the receptionist's silence — it now answers property questions on real data, and the facts table is live in production as of this session. B5's queue is deployed and consuming, and B4's checklist is down to one line: move watcher-api off the free plan so a cold start doesn't read to Meta as a timeout. The single positive proof still worth watching for is the first real inbound message landing a row in messages.

Track A — Make it run. [COMPLETE — ALL SIX DONE]

Configuration, a database, a model, an endpoint, continuity and a sender. The track that turned a library into a running receptionist is finished; what follows is hosting it safely.

#	Work item	Urgency	Ease	Days	Status
A1	Configuration layer	DONE	Easy	0.5	DONE — core/config.py; every variable in .env.example typed, required per subsystem, placeholders read as absence, secrets as SecretStr. G3 added one field: TENANT_TIMEZONE, and it is the one defaulted value this layer validates eagerly
A2	DB engine and session	DONE	Easy	0.5	DONE — db/engine.py; engine, sessionmaker, session scope. Transaction-mode pooling assumed by default (NullPool + prepare_threshold=None), DATABASE_POOL_MODE opts out; a bare postgresql:// URI is rewritten onto psycopg3; alembic connects the same way
A3	Concrete LLM providers	DONE	Moderate	1.5	DONE — Anthropic + OpenAI on the existing seam; system block marked cacheable; per-call usage reported, so cost is measurable
A4	Composition root	DONE	Moderate	1.0	DONE — apps/api/main.py, run as uvicorn apps.api.main:create_application --factory. Also the five database port implementations create_app needed and a process-level worker pool, so the webhook still answers before classification runs
A5	Wire continuity + excise v1 filer	DONE	Moderate	2.25	DONE — a ConversationStore in the message path: the inbound turn is recorded, the active task is loaded and saved back, the reply joins the transcript. process() is async; classifications is populated with measured latency and the prompt version; the rules/destinations threading is gone (D24), tables retained
A6	Outbound sender	DONE	Moderate	1.0	DONE — WhatsAppSender on the ChannelSender seam, quick replies rendered within the channel's limits, bounded retries. The credential fields moved behind channels/config.py and the last KNOWN_LEAKS entry is gone with them

What Track A leaves behind. Items 2.1 and 2.2 were genuinely written — the tables, the repository, the task state machine, the reply path — and the line joining them never was. A5 wrote that line and A6 wrote the wire. G3 has now made answering safe, so the milestone is no longer “holds a conversation” but **holds a conversation it is allowed to be having**. As of 2.4 it also has something true to say. What is between this and a real guest is B4.

Track B — Host it. [Supabase + Render]

Stack locked 15 August 2026. B2 landed before a second client exists. B5 is fully done as of this session — the worker service now exists; B4 is done bar the paid-plan upgrade — see their notes.

#	Work item	Urgency	Ease	Days	Status
B1	Supabase project and migrations	DONE	Easy	0	DONE — watcher-prod in eu-central-1, migrations applied and stamped, one enabled channel_configs row. Its external_id was a placeholder until session 10, when it was replaced with the real 16-digit Meta phone-number id by a live UPDATE, verified by reading the row back
B2	Row-Level Security	DONE	Moderate	0	DONE — migration 004: enabled and forced on all 20 tables, policies on app.current_tenant, and a watcher_app role that cannot bypass them (Supabase's postgres role can, which is the trap). Verified cross-tenant on the live database
B3	Dockerfile and Render service	DONE	Easy	0	DONE — apps/api/Dockerfile, which activates the cd.yml image job, and the service is live in Frankfurt. WHATSAPP_ACCESS_TOKEN and WHATSAPP_PHONE_NUMBER_ID were set on Render in session 10, confirmed by a clean redeploy — not yet by an actual outbound send. Still on the free plan
B4	Webhook subscription and a warm instance	NOW	Easy	0.25	DONE bar the plan upgrade — the Meta webhook subscription and the operator number (CONTROL_CHAT_PHONE_E164) are completed operator-side, closing the two items session 11's direct check had re-opened. What is independently confirmed this session: webhook.automera.co is the service's primary domain (Render's deploy log names it), and the app starts clean with META_APP_SECRET and META_WEBHOOK_VERIFY_TOKEN both set. One operational sub-item remains: watcher-api is still on the Render free plan (confirmed via the Render MCP this session) — the free one sleeps after ~15 minutes and a 30–60s cold start reads to Meta as a timeout, so the upgrade is what makes the warm-instance half of this item real. Note for whoever picks this up: at check time the messages table still had zero rows and /webhook zero recorded hits, so the first real inbound message is still the positive proof to watch for
B5	Durable queue (Redis + arq)	DONE	Moderate	0	DONE — RedisClassificationQueue as a fourth ClassificationQueue transport, apps/api/worker.py as its arq consumer, and orchestration/composition.build_consumer as the one graph-builder both the API and the worker call, so the knowledge base (2.4) and everything else can't drift between them. Reconciled onto main after 2.4 landed. The provisioning half, open at the end of session 11, is now closed: watcher-redis (a Render Key Value instance in Frankfurt) and watcher-worker — a paid Background Worker running arq apps.api.worker.WorkerSettings — both exist and are live, the worker's own logs confirming it started for one function (consume_message) and connected to Redis. Verified via the Render MCP this session

Runs in parallel — start today. P1 pick the first client (NOW, blocks 3.1) • P2 PMS sandbox keys (HIGH, 0.5d) • P3 file the platform business verification (HIGH, 1 hr) • P4 Graphify as a build aid (LOW, optional). Founder time, excluded from the total — and P2 and P3 are the two that make Track B's week land on time.

P5 is new and it is 30 minutes: widen the emergency trigger phrases in intents.yaml. It changes no code, moves no baseline, and it is the cheapest safety work on the board.

Track 2 — Make it a receptionist. [COMPLETE]

2.7 and 2.8 both landed this session — the prompt is measured (0.98, v3) and a client can be an agency. Track 2 is done.

#	Work item	Urgency	Ease	Days	Status
2.1	Conversations, tasks and slot filling	HIGH	Hard	spent	WIRED (A5) — 2.0d spent here, the wiring priced in A5. Slot extraction is the one part still missing: the model emits no slots, so a task fills only by the clarifying-turn budget expiring. That is a prompt change and a golden-set change
2.2	The reply path	HIGH	Moderate	spent	WIRED (A6) — 1.5d spent here, the sender priced in A6
2.3	Autonomy gate	DONE	Easy	1.0	DONE — RECEPTIONIST_REPLY as the fourth RoutingAction. G3 added a fifth, EMERGENCY, which is deliberately not a variety of handoff: it was never classified
2.4	Knowledge base	DONE	Moderate	spent	DONE — a facts table (migration 005) behind row-level security, matched against a guest's own words with a stopword-stripped rapidfuzz score (D35), a sensitive flag a fact can carry (D36), and answer_from_knowledge actually dispatched from the receptionist for the first time — every other terminal_tool still gets the old placeholder (roadmap 3.1). Door codes, key codes and unit numbers are excluded from the table by design, not merely unmarked. Tested against a curated, committed export of one real property. 486 → 506 tests; decisions D35–D37
2.5	Prompt v2 and rewrite the golden set	DONE	Moderate	1.0	DONE — golden set 8 → 50 across all 19 intents
2.6	Intent taxonomy unification	DONE	Easy	0.5	DONE — 19 vocabulary-aligned intents
2.7	Re-record the eval under prompt v3	DONE	Easy	0.5	DONE — the fixtures were re-recorded live against claude-haiku-4-5 under prompt v3 (scripts/record_fixtures.py, first pass only, escalation pinned off) and the baseline moved from v2's 0.88 to a measured 0.98 (55/56). Six Franco-Arabic cases were added to the golden set in the same pass (50 → 56); the model handled all six. The one miss is a borderline general_info-vs-property_question case ('where is the nearest supermarket') the model called property_question at 0.78 — below the escalation threshold, so production re-runs it on the larger model rather than acting on the cheap guess. scripts/measure_prompt.py also settled the two page-9 numbers: the system prompt is 21,777 characters, clear of Haiku 4.5's 4,096-token cache floor. G3 does not move the number: an emergency bypasses the classifier
2.8	Many properties per client	DONE	Moderate	1.0	DONE (code) — a properties table (migration 006) behind the same forced RLS every tenant table carries, a nullable facts.property_id that scopes a fact to one unit (NULL = tenant-wide, the common case), and message → property resolution (core/property.py): an explicit endpoint hint wins, else a single-property tenant resolves to its one unit, else tenant-wide facts only — a guest on a shared number is never answered from the wrong unit's sheet. The knowledge lookup returns the resolved property's facts plus the tenant-wide ones. 521 → 535 tests. Migration 006 applies to production on deploy, the same step 005 needed (D5's editing UI carries the door/key/unit-code exclusion forward when it lands)

2.4's sensitivity flag is not G1. A fact marked sensitive is withheld from an unverified guest — treated exactly like a genuine “I don't know” — but that is one tool refusing to guess, not the reply-path-wide disclosure gate G1 still has to build. And a door code, a key box code or a unit number does not go into the facts table at all, sensitive or not: `intents.yaml` forbids `answer_from_knowledge` from ever giving one out, and that is a table an operator (or D5, once it exists) populates by hand.

The eval now measures the prompt. For four revisions the gate replayed v2 fixtures and reported 88% whatever the prompt said. 2.7 re-recorded the fixtures live under prompt v3 (claude-haiku-4-5, first pass only), and the number moved to a **measured 0.98** (55/56) — the baseline in `baseline.json` is now v3's, not v2's. The gate still replays the recorded outputs for determinism (no key in CI); what changed is that the outputs are v3's. Six Franco-Arabic cases were added to the golden set in the same pass.

Track G — Receptionist guardrails 4.0d

Not new features. Every one of these enforces a rule that intents.yaml already states and schema.py already validates, and that nothing in apps/ honours — the vocabulary is obeyed when composing a reply and ignored when deciding whether to send one. **G3 is done, and with it this track leaves the critical path.**

#	Work item	Urgency	Ease	Days	Status
G3	Emergency detection and the alert path	DONE	Moderate	0	DONE — core/emergency.py matches the declared triggers before the classifier and after the media pipeline; the guest is answered immediately in English and Arabic; the item is filed NEEDS_REVIEW at the top band; an OperatorAlerter seam reaches a person and reports which channel it used, because the vocabulary asks for a phone call nothing wired can place. An emergency is never classified. 417 → 486 tests; decisions D30–D34; docs/specs/g3-emergency-path.md
G1	Sensitivity and disclosure gate	HIGH	Moderate	1.5	Facts now carry a sensitivity flag and answer_from_knowledge refuses one to an unverified guest (2.4) — but that is one tool, not the reply-path-wide gate: money and owner matters still reach a confident model with no ceiling, and the identity-verified flag on conversations is read in exactly one place rather than everywhere the vocabulary implies it should be
G2	Autonomy ceilings in the reply path	HIGH	Easy	1.0	Money and owner matters must reach a human before confidence is consulted. The gate exists (2.3) and the vocabulary declares per-intent ceilings; what is missing is the enforcement that a ceiling cannot be raised by a confident model
G4	Anti-loop and abuse controls	MED	Moderate	1.5	The clarifying-turn budget bounds one task. It does not bound a guest who sends forty messages, a channel that redelivers, or two tasks that hand off to each other. G3 added one instance of this deliberately: five messages about the same fire raise five alerts, which is the right default for a safety path and the wrong one for an operator's phone

What G3 settled

The ordering. intents.yaml has said “checked before intent, before confidence, before anything” since item 0.3, and that is now literally where the check is. A model is a network round trip that can be slow, wrong or down, and none of that may stand between a guest saying *smell of gas* and an operator

What it left open, on purpose

phone_call_to_operator is not satisfied. The alerter delivers a message and says so — AlertOutcome.channel beside EmergencyAlert.requested_channel, false today, logged per emergency and once at startup. A voice alerter is a new implementation of the same protocol and needs the same dependency as the phone channel

What is the operator's to do

Widen the trigger phrases. The detector matches what is written down and does not paraphrase, which is what makes intents.yaml reviewable by the person carrying the consequences — and it means “I smell gas” matches neither declared gas phrase. One line of YAML; it touches neither the prompt nor the eval baseline

Track D — The control page 13.75d

The largest remaining track, and the one whose shape is most often misread. D2 is three backend days inside an item everyone reads as frontend work — the views cannot be built against endpoints that do not exist. Views follow DESIGN-SPEC §8, re-ordered for the receptionist: what the product does now is hold conversations, not file messages.

#	Work item	Urgency	Ease	Days	Status
D0	Frontend scaffold and design tokens	MED	Easy	1.0	One CSS file and no package.json today. Tokens, type scale and the confidence chip come straight from DESIGN-SPEC §2–§6; the chip is the signature component and the bands are semantic, never reused
D1	Auth and tenant binding	MED	Moderate	1.25	Clerk (D2-a), one auth org to one tenant. This is where the RLS session GUC (B2) gets its value from an authenticated principal rather than from the message
D2	The REST API behind the page	HIGH	Hard	3.0	~25 endpoints, none written. Inbox list and detail, confirm/correct/route, conversations and turns, knowledge CRUD, sources, destinations, rules, eval. The hidden half of this track and the item most likely to be cut by accident
D3	Inbox view	HIGH	Hard	2.0	The triage queue and the product's centre of gravity. Two-pane desktop, single-column mobile, optimistic confirm. Auto-handled items still appear, marked, for audit — and emergencies arrive here at the top band, filed for review with the trigger named
D4	Conversations view	HIGH	Moderate	1.5	The thread, its turns, the task in flight and its slots. Where a human sees why the receptionist asked what it asked — and, after G3, why it stopped asking
D5	Knowledge view	MED	Moderate	1.0	The editor for 2.4's facts, including the sensitivity flags G1 enforces. Without it, changing a check-in time is a database write
D6	Sources view	LOW	Easy	0.75	Opt-out model (addendum §4): a settings page listing threads with a mute toggle, plus the first-run exclusion pass. Deliberately not a workflow
D7	Destinations and recipes	MED	Moderate	1.0	Sheets/webhook config and field mapping. The tables and the engine survived D24 for exactly this: routing a message deliberately, by a human
D8	Rules builder	MED	Moderate	1.0	The condition/action builder over rules. SqlAlchemyRulesProvider and the evaluator are already written and already tested — this is the surface for them
D9	Admin / eval dashboard	LOW	Easy	0.75	Founder-only, role-gated, visually distinct: accuracy, per-language breakdown, calibration, confusion matrix, and per-tenant model spend
D10	Arabic, RTL and the accessibility pass	MED	Moderate	0.5	Not cosmetic for this market. Bidi text, mirrored layout, and the §10 baseline (focus order, contrast, target sizes) applied once across the views rather than per view

Track E — What makes it sellable 4.5d

None of this is visible to a guest and none of it can be added after the first paying tenant without a migration. Sequenced after P1 because onboarding shape depends on who the first client is.

#	Work item	Urgency	Ease	Days	Status
E1	Tenant onboarding and per-tenant secrets	MED	Moderate	1.25	Creating a tenant is currently an INSERT by hand. Includes envelope encryption for WABA tokens and destination credentials (addendum §3: never plaintext) — the reason channel_configs.config is the one table B2 grants a pre-tenant read on
E2	Usage metering	MED	Easy	0.75	usage_events exists, is indexed, and is written to by nothing. Messages, model tokens and ASR minutes per tenant per period — the input to both billing and the soft cap
E3	Billing and plan limits	MED	Moderate	1.0	Plans, the soft cap and what happens at it. The number itself is still a founder decision (DECISIONS.md, still open) — the mechanism is not
E4	Observability	HIGH	Easy	1.0	Structured logs with tenant and message ids, error tracking, and per-tenant model spend. Today a failed classification is a line in a Render log and nothing else — and so is an undelivered emergency alert, which is the line that most needs to page someone
E5	Retention, residency and subject deletion	MED	Moderate	0.5	Addendum §14: a configurable retention default, hard deletion of raw bodies and media after it, and a “delete everything for this contact” operation. GCC regulated buyers ask for this in the first meeting

Track 3 — Integration and measurement 5.5d

Base360.ai is ruled out as a partner: their product is substantially ours and their client base is closed to us. Hostaway, Guesty and Cloudbeds all publish APIs, so the capability is available without the strategic cost.

#	Work item	Urgency	Ease	Days	Status
3.1	PropertySystemPort and the first adapter	MED	Moderate	2.5	Build the port, not a Hostaway integration — the vendor-neutral routes and schemas already landed (PR #14). Blocked on P1: which PMS the first client runs decides which adapter gets written. Cache facts; never cache availability
3.2	End to end on a real number, then measure	HIGH	Moderate	1.0	The point at which the eval number stops being a replay of recorded fixtures and starts being this product's accuracy. G3 is no longer on its list of prerequisites ; what remains is B4 and the send credentials
3.3	Live availability and write-back	MED	Hard	2.0	Reading a calendar is half of it; holding a booking is the half that makes the receptionist worth paying for, and the half that must never act on a stale cache

New this revision: what G3 settled, and what it left open

The ordering, literally	The emergency check runs in Orchestrator.process, after the media pipeline and before the classifier . After media, because a voice note saying there is a fire has no text until it is transcribed. Before the classifier, because a model is a network round trip that can be slow, wrong or down. Asserted twice: against a classifier that raises if it is called at all, and end to end through the assembled graph
Two scripts, two rules	Arabic matches as a substring, because Arabic attaches its article to the front of the word — ■■■■ is ■■■■■■ with the article, and a boundary test there invents false negatives on the most ordinary way to write the sentence. Latin matches on word boundaries, because <i>fire</i> as a substring fires on “fireplace”. Digits count as letters, so Franco-Arabic works and 7ari2 is a word rather than a fragment
One trigger needs a clock	locked_out_at_night carries only_between, and the vocabulary’s own note says why: locked out at 2pm is a support request, at 2am it is a person on a street. The window is read in TenantPolicy.timezone — the guest’s clock, not the container’s. Dubai and Cairo are an hour or two apart, which is an hour of the window either side of midnight
Reply and alert, concurrently	Both halves of the exchange are recorded first, then dispatched together. Sequentially, one waits on the other’s retries, and neither “the guest hears nothing for ten seconds while we retry the alert” nor the reverse is an acceptable way to spend that time. The alerter never raises — a failed alert must not cost the guest the immediate reply
The job in flight survives	An emergency reply belongs to the transcript and to no task, so ConversationStore.record_reply takes an optional one. The active job is left exactly as it was rather than abandoned: a guest with a gas leak may well return to their booking question, the conversation belongs to a person either way, and discarding what the receptionist collected only makes the resumed conversation worse
Still not written	Slot extraction . The receptionist receives an empty slot dict because ClassificationResult has no slots field; adding one is a prompt change and a golden-set change, which is item 2.x. The declared phrases are narrow — the detector does not paraphrase, by design, so widening them is the operator’s edit and it is the first thing to do before a real guest can reach the number

Two numbers 2.7 settled, and the tool that keeps them settled

1. The prompt’s real token count	The system prompt is 21,777 characters (measured, not estimated), ~5.4k tokens by the characters÷4 rule — clear of Haiku 4.5’s 4,096-token cache floor, so the cacheable prefix does cache. scripts/measure_prompt.py prints the exact count.tokens figure and warns if the prefix ever falls below the floor (a shrinking vocabulary would stop caching silently: no error, just a larger bill). Sonnet 5’s floor is 1,024, so escalation is not exposed either way
2. Cost per message	scripts/measure_prompt.py computes it from the measured token count and Haiku 4.5 pricing (\$1/\$5 per MTok, cache read ~\$0.10): a cold message pays full input on the whole prefix, a warm one pays cache-read, and the tool prints both. G3 makes one class of message free: an emergency never reaches a model

How the total reconciles

A 0 + B 0.25 + D 13.75 + E 4.5 + G 4.0 + Track 2 (complete) 0 + (3.1, 3.2, 3.3) 5.5 = **28.0** — 2.7 and 2.8 delivered this session close Track 2

Rows 2.1, 2.2, 2.4, 2.7, 2.8 and B5 show “spent” or 0 rather than a number because their days are already delivered and their wiring was priced in A5, A6 and the sessions that built them — adding them would double-count. B4’s remaining 0.25 is operational (a plan upgrade), not engineering; the subscription, the operator number and the worker are done. P1–P5 is excluded: founder and operator time, not engineering. Counting scope loosely is what produced the 5.75 figure that v2.1 replaced, so the arithmetic is stated rather than implied.

The dates

Milestone	v2.8 said	Now	Status
M1 — answers a real message, safely	~4 days	~0.25 days	NEXT — B4's last item alone. It is deployed, it answers, as of G3 it answers <i>safely</i> , and as of 2.4 it has something true to say. The webhook subscription, the operator number and the worker are all reported or verified done this session; what remains is the paid Render upgrade. The positive proof still worth watching for is the first real inbound message landing a row in messages
M2 — you can watch and correct it	+13.75 days	+13.75 days	PENDING — Track D, six receptionist views
M3 — sellable	+16.5 days	+16.5 days	PENDING — Tracks E, 3 and the rest of G. Blocked on P1
Phone answering	after M3	after M3	PENDING — the speech-to-text seam already exists, and G3 added a second reason to want a voice channel: <code>phone_call_to_operator</code>

Read M1 carefully, because its shape changed rather than only its number. In v2.8 the first milestone contained a hazard: a deployed receptionist that answered a gas leak with a note about maintenance. As of v2.9 that was closed; as of 2.4 it also has something true to say. As of this revision the checklist is down to its last line — a paid Render plan — with the subscription, the operator number and the worker reported or verified done. The remaining risk on the path to a pilot is **operational**, and operational risk is the kind a checklist closes.

What would actually move these dates

The risks v2.3 named, re-scored against what session 9 learned.

A receptionist that answers but cannot recognise an emergency	CLOSED (G3). It was the most serious item on this page in v2.8 and it is now a detector, an immediate bilingual reply, an operator alert and 69 tests. What replaces it is narrower and named below
The trigger phrases are narrower than they read	NEW, and the successor to the item above. The detector matches phrases an operator wrote down and deliberately does not paraphrase — that is what makes intents.yaml reviewable by the person carrying the consequences. The cost is that "I smell gas" matches neither declared gas phrase. A test asserts this so it cannot be forgotten, and the fix is one line of YAML that touches no code
Nobody is told, because nobody is configured	Reported done. Without CONTROL_CHAT_PHONE_E164 the alert is a log line; the process warns at startup and refuses nothing, which is the right trade for a service that still ingests and files — and the wrong state to point a guest's number at. The operator reports it set this session; the deployed worker still logged the startup warning at last check, so a redeploy confirming the warning is gone is the positive proof
The alert is not the phone call the vocabulary asks for	NEW, bounded and reported. A message can be missed in a way a ringing phone cannot. The gap is visible in AlertOutcome rather than smoothed over, and closing it needs a voice provider — the same dependency as the phone channel
Supabase connection pooling	Unmeasured. A2 assumes the transaction pooler and is safe there by default. What is not known is the cost of that safety — NullPool trades a handshake per checkout for correctness. Measure against the real pooler now that a service is running
The deployed database path is unproven	Unchanged. SQLAlchemy connects lazily, so startup proves nothing and the first tenant query is the test. If the first real message fails on DNS or auth, check the pooler host against Supabase → Connect before looking anywhere else
RLS retrofitted late	CLOSED (B2). Kept on the page for one revision because the trap it names — connecting as a role with BYPASSRLS — is a configuration mistake, not a code one, and can be reintroduced by a single environment variable
D2 is invisible in its own track title	Unchanged, and now the most likely accidental cut on the board. Three of Track D's days are backend
Cost per message	Unmeasured, not unknown. A3 made it measurable and turned prompt caching on. Measure before quoting a price
PMS API access	Unchanged. Docs are public, but sandbox approval and rate limits are theirs to grant. Start P2 today
Roadmap artifacts disagree	RE-OPENED, then closed. v2.8 declared this shut, but its generator was never committed — the repository could only produce v2.5, so the PDF everyone was reading had no source. v2.9 re-derives v2.8 and folds session 9 into it. The rule stands: edit docs/make_roadmap.py and re-run it; never hand-edit the PDF
Two messages arriving at once	Infrastructure now in place. B5's durable queue and its arq worker (watcher-worker) are deployed and consuming as of this session. Ordering only actually changes once REDIS_URL is set on watcher-api to switch it to the producer branch — a deliberate step, taken after the worker is confirmed consuming, so a message can't be accepted and then never replied to. Until then the API still runs on the in-process pool it has since A4
Webhook subscription was reported done and was not	Reported closed operator-side; positive proof still pending. Session 11 checked it directly — zero /webhook requests ever, zero messages rows — against a genuinely correct infrastructure (custom domain, both Meta credentials). This session the operator reports the subscription and the operator number completed. The standing lesson holds: this line closes on a positive check — a real handshake or a real message landing a row in messages — not on a status carried forward. At check time both were still empty, so the first inbound message is the thing to watch for

If you do only one thing next session: confirm B4 with a real message, then upgrade the Render plan.

Not D, not G1, and not 2.4 or B5 — all done, and B5's worker is now live. The receptionist answers, it is deployed, it knows when to stop and fetch a person, it has something true to say, and the facts table is live in production. The webhook subscription and the operator number are reported done operator-side; the one engineering-adjacent item left is moving watcher-api off the free plan so a cold start does not read to Meta as a timeout. **The positive proof to insist on** is still the first real WhatsApp message landing a row in the messages table (or a POST /webhook 200 in Render's logs) — at this session's check both were empty, so treat B4 as closed only once one appears. **And cheaply, spend thirty minutes widening the emergency trigger phrases** — it is the operator's edit, it changes no code, and it is the cheapest safety on the board.

Session log

Session	Delivered	Tests	Days
1	PR #13 — items 1.1, 1.3, 2.1, 2.2, 2.3	228 → 248	6.5
2	PR #14 and #15 — items 2.5, 2.6	248 → 259	1.5
3	PR #16 — prompt v3, unplanned	259 → 277	~0.5
4	Code audit plus a design review. No application code changed. Five unpriced tracks added; 2.1/2.2 downgraded to part-built; emergency detection found unwired; the control page rebuilt around six receptionist views. Decisions D14–D26	277	—
5	PR #17 — items A1, A3. Config layer over all ~20 env vars; Anthropic + OpenAI providers with prompt caching and per-call usage; D8-a re-pinned to the Claude 5 family	277 → 325	2.0
6	PR #18 — items A2, A4. Engine and session scope, pooler-safe by default; the composition root and the five database port implementations it needed; a process-level worker pool so the webhook still answers before classification; alembic aligned to the same connection path	325 → 375	1.5
7	PR #19 — items A5, A6. Track A complete. Conversation and task continuity in the message path; the clarifying-turn budget honoured; classifications populated with measured telemetry; process() async; the v1 rules/destinations filer excised (D24); the outbound sender, and the channel credentials moved behind channels/, which emptied KNOWN_LEAKS and closed item 1.1	375 → 406	3.25
8	PR #20 — items B1, B2, B3. Supabase project provisioned, migrated and stamped, with the channel_configs row tenant resolution raises without; migration 004 — RLS enabled and forced on all 20 tables behind a watcher_app role that cannot bypass it, a per-transaction tenant GUC carried by every adapter, and one narrow SELECT-only exception for the endpoint lookup that runs before a tenant is known; the container image, whose first build exposed a missing package data file and a removed Starlette API; deployed to Render, Frankfurt, and serving	406 → 417	2.75
9	PR #21 — item G3. The emergency path: a detector over the declared triggers, matched in Arabic, Latin and Franco-Arabic and placed before the classifier; an immediate bilingual reply; an operator alert on a seam in the core with its implementation in channels/, reporting which channel it used because the vocabulary asks for a phone call nothing wired can place; a per-tenant timezone for the one trigger with a clock, validated at startup. An emergency is never classified. Decisions D30–D34	417 → 486	1.5
10	Item 2.4. A facts table (migration 005) behind row-level security; core/knowledge.py's stopword-stripped rapidfuzz matching (D35); a sensitivity flag a fact can carry, enforced narrowly — an unverified match to one is treated exactly like no match (D36); answer_from_knowledge dispatched from the receptionist for the first time, replacing a placeholder reply that had been silently wrong for every informational intent; a one-off import script and a committed, curated fixture from one real property, exercised end to end in test_knowledge_integration.py. REGISTRY wired via configure_knowledge, not threaded through Orchestrator (D37); an autouse fixture keeps that process-global state from leaking between tests. Decisions D35–D37	486 → 506	2.0
11	Item B5, reconciled with 2.4; B4 re-investigated. Merged the durable arq/Redis queue built on a branch that had forked before 2.4 (RedisClassificationQueue, apps/api/worker.py, orchestration/composition.build_consumer as one graph shared by the API and the worker) onto main after 2.4's merge, moving 2.4's configure_knowledge call into the shared graph-builder so the arq worker gets the knowledge base too. Decisions D38–D40. Separately, re-checked B4's "webhook subscribed" claim directly against Render's request logs and the Supabase messages table rather than repeating it: the custom domain and both Meta credentials are genuinely correct, but zero requests to /webhook have ever been recorded and zero messages exist — the subscription step was not actually done, or was undone. Provisioned watcher-redis, a free-plan Render Key Value instance, on request; the worker service is the one piece left unprovisioned — it needs Render's Background Worker type, which this session's tooling can only create as a (wrong) Web Service, plus secrets only the operator can copy over, so REDIS_URL stays unset on watcher-api until it exists	506 → 521	1.0

12	B4 and B5 closed on the roadmap; migration 005 applied to production. Verified the live systems rather than the handoff: watcher-worker — the arq Background Worker that was session 11's one open piece — now exists as a paid Render service, deployed and consuming (logs confirm it started for consume_message and connected to Redis), so B5 is fully done. B4's webhook subscription and operator number are taken as completed operator-side, leaving only the paid watcher-api upgrade — B4 down to 0.25d. Found production schema still at 004 with migration 005 (2.4's facts table) never applied, and applied it: the live schema is now 005_facts, RLS forced, tenant-isolation policy and watcher_app grant in place. Then delivered item 2.8 — a properties table (migration 006), a nullable facts.property_id scoping a fact to one unit, and message → property resolution (core/property.py) — and item 2.7 : re-recorded the eval live under prompt v3 (scripts/record_fixtures.py, first-pass claude-haiku-4-5), moving the baseline from v2's 0.88 to a measured 0.98 (55/56), and added six Franco-Arabic golden cases. scripts/measure_prompt.py settled the prompt size (21,777 chars). Track 2 complete. Decisions D41–D43	521 → 535	1.5
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Cumulative: 24.0 engineering days delivered. ~28.0 remaining. Twelve sessions — session 12 closed B4/B5's infrastructure, applied migration 005, and delivered 2.7 and 2.8, completing Track 2 — and the observed rate holds at 1–3 days each.